Hein Zarni Aung

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EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Algorithms, Statistics & Probability

INTERNSHIPS

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, and custom date/time picker integration to improve user engagement.

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

PROJECTS

TuneAcademy React, FastAPI, Essentia, ElevenLabs, Firestore

- Built the AI/audio-analysis pipeline for an app matching music students with instructors, using Essentia and NumPy to extract pitch, rhythm, and tone features and rule-based scoring algorithms to detect technical weaknesses within a 36-hour hackathon build window.
- Built a React/Tailwind frontend backed by a Python/FastAPI backend, Cloud Firestore database, and ElevenLabs-generated AI voice feedback reports, with a CI/CD pipeline via GitHub Actions to lint, test, and deploy.

Diary Handwriting Analysis Python, PyTorch, Hugging Face Transformers, OpenCV

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV preprocessing pipeline, reaching ~87% reliable transcription, then pivoted to DANIEL, a segmentation-free architecture for joint transcription and named-entity extraction to surface mood and topic patterns across diary entries.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University Sep 2025 – Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time at scale, scraping keywords from the raw HTML, CSS, JS, and URLs of newly created sites to flag domains impersonating EZ-Pass tolling services.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage hundreds of flagged sites, designing keyword- and heuristic-based detection logic to separate real fraud from noise.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations; created original midterm review materials adopted as the basis for later semesters and hosted review sessions reaching 100+ students.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C

Frontend: React, React Native, Next.js

Backend: FastAPI, Node.js, Express.js

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, Supabase, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium